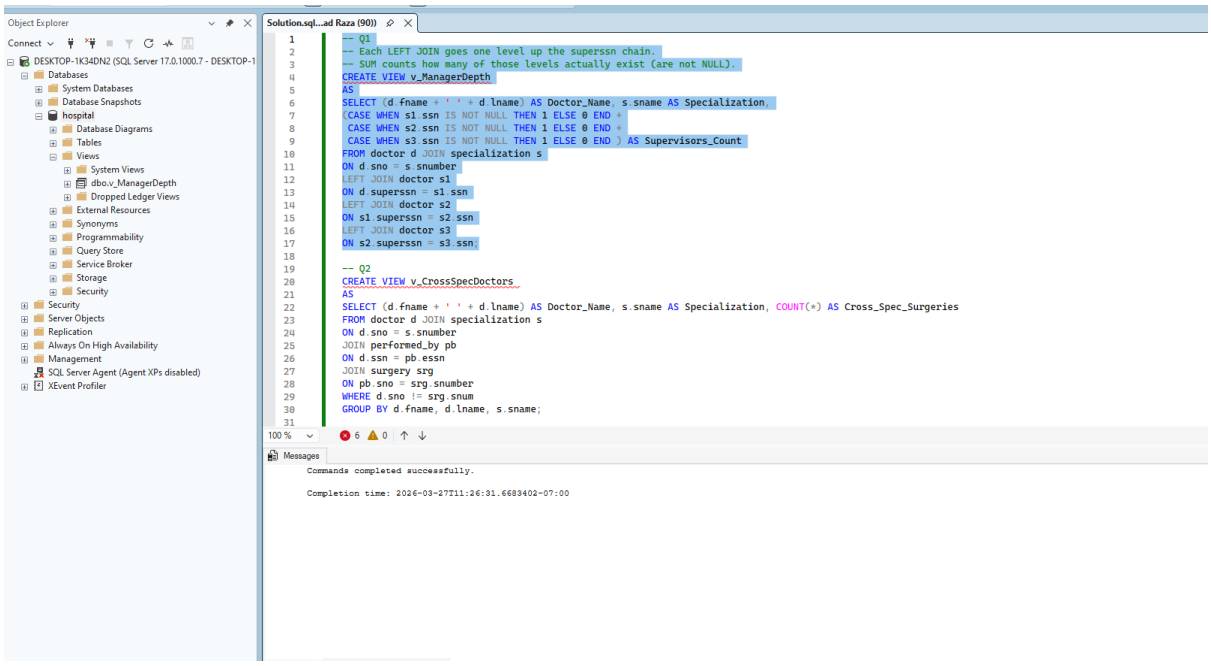
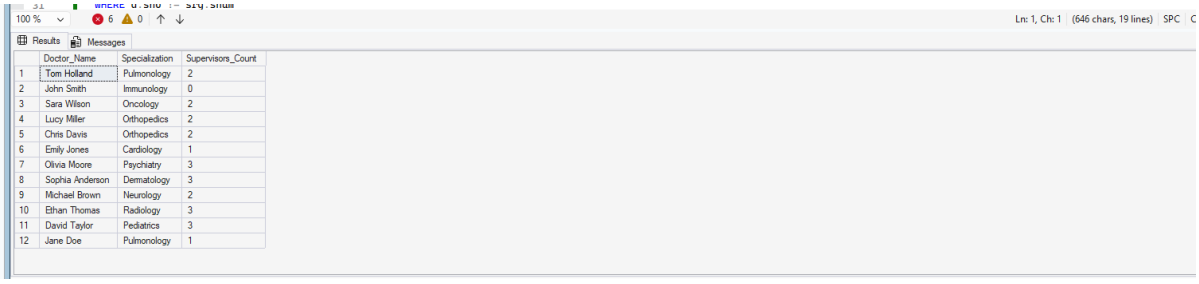
# Q1




| | Doctor_Name | Specialization | Supervisors_Count |
|---|---|---|---|
| 1 | Tom Holland | Pulmonology | 2 |
| 2 | John Smith | Immunology | 0 |
| 3 | Sara Wilson | Oncology | 2 |
| 4 | Lucy Miller | Orthopedics | 2 |
| 5 | Chris Davis | Orthopedics | 2 |
| 6 | Emily Jones | Cardiology | 1 |
| 7 | Olivia Moore | Psychiatry | 3 |
| 8 | Sophia Anderson | Dermatology | 3 |
| 9 | Michael Brown | Neurology | 2 |
| 10 | Ethan Thomas | Radiology | 3 |
| 11 | David Taylor | Pediatrics | 3 |
| 12 | Jane Doe | Pulmonology | 1 |

Q2

```
FROM doctor d JOIN specialization s
ON d.sno = s.snumber
LEFT JOIN doctor s1
ON d.superssn = s1.ssn
LEFT JOIN doctor s2
ON s1.superssn = s2.ssn
LEFT JOIN doctor s3
ON s2.superssn = s3.ssn;

SELECT * FROM v_ManagerDepth;

-- Q2
CREATE VIEW v_CrossSpecDoctors
AS
SELECT (d.fname + ' ' + d.lname) AS Doctor_Name, s.sname AS Specialization, COUNT(*) AS Cross_Spec_Surgeries
FROM doctor d JOIN specialization s
ON d.sno = s.snumber
JOIN performed_by pb
ON d.ssn = pb.essn
JOIN surgery srg
ON pb.sno = srg.snumber
WHERE d.sno != srg.snum
GROUP BY d.fname, d.lname, s.sname

SELECT * FROM v_CrossSpecDoctors;
-- Q3
CREATE VIEW v_InvalidSurgeryLocations
AS
SELECT srg.sname AS Surgery_Name, srg.slocation AS Location, s.sname AS Specialization_Name,
CASE WHEN EXISTS (SELECT 1 FROM spec_locations sl
WHERE sl.snumber = srg.snum AND sl.slocation = srg.slocation) THEN 'VALID' ELSE 'INVALID' END AS Status
```

Messages

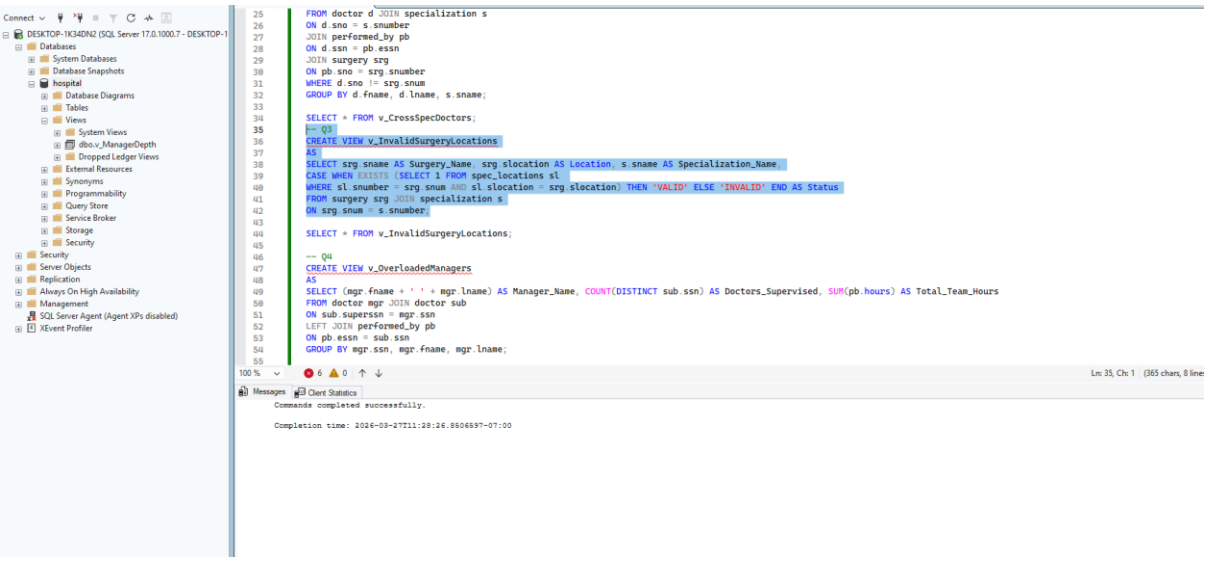
```
Commands completed successfully.

Completion time: 2026-03-27T11:27:49.5881988-07:00
```

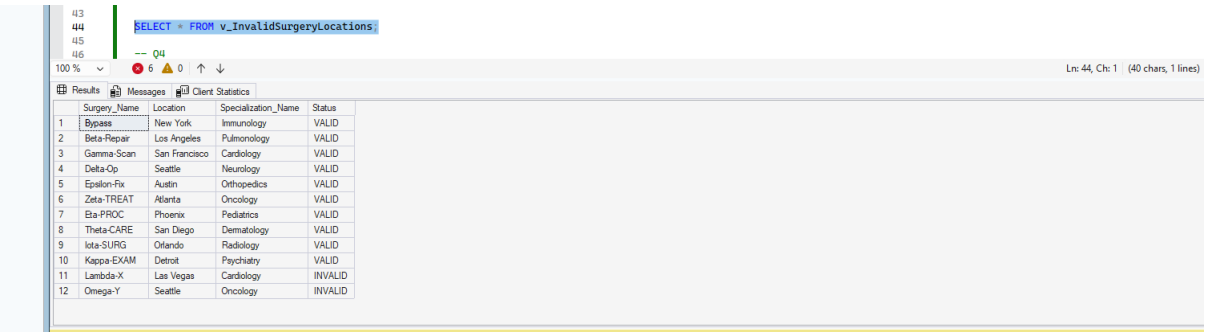
```
SELECT * FROM v_CrossSpecDoctors;
-- Q3
CREATE VIEW v_InvalidSurgeryLocations
AS
```

Results | Messages | Client Statistics

| | Doctor_Name | Specialization | Cross_Spec_Surgeries |
|---|---|---|---|
| 1 | David Taylor | Pediatrics | 1 |
| 2 | Jane Doe | Pulmonology | 1 |
| 3 | John Smith | Immunology | 1 |

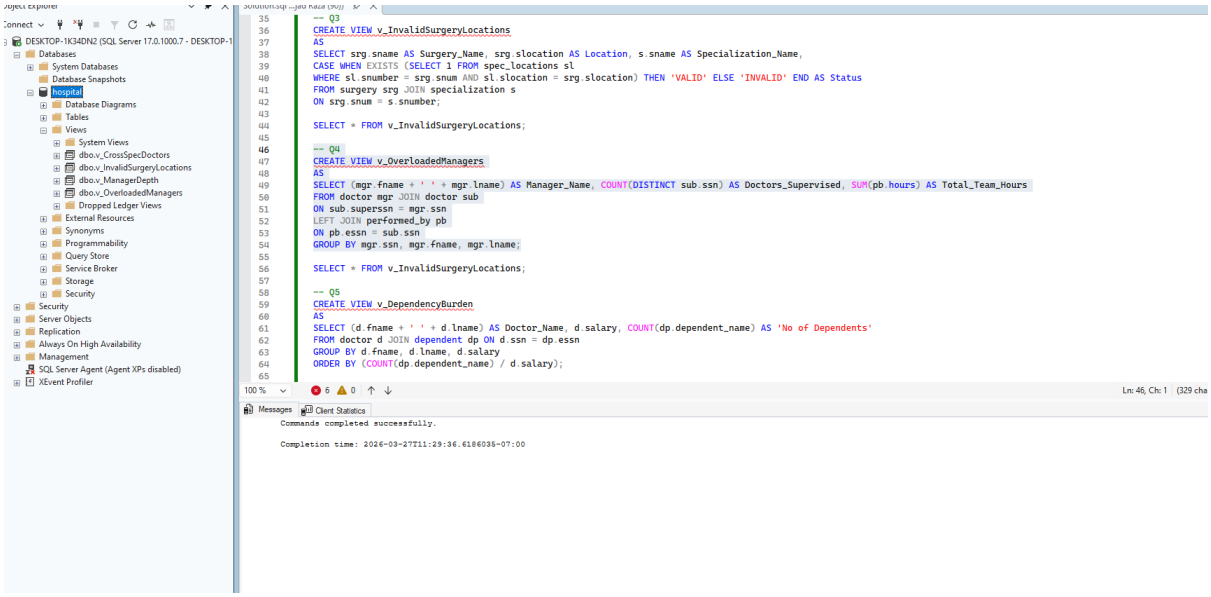
# Q3

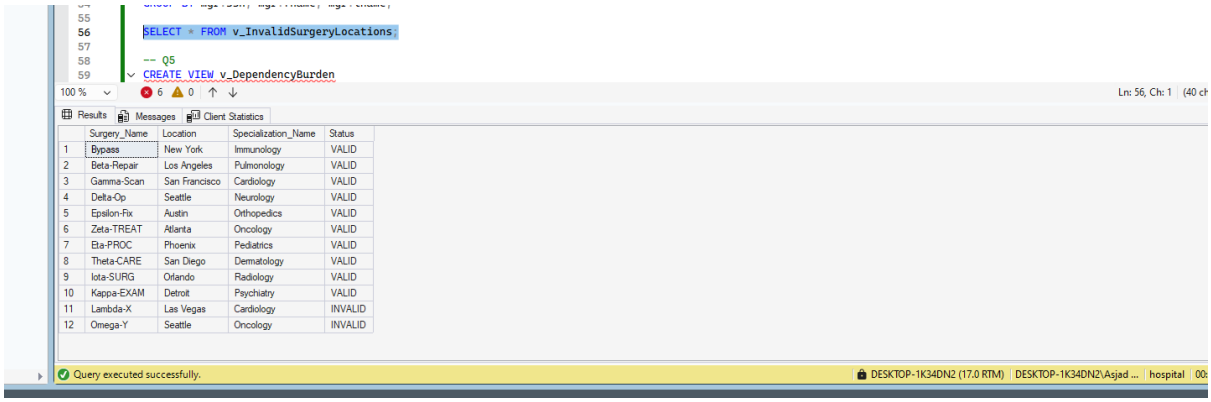



| | Surgery_Name | Location | Specialization_Name | Status |
|---|---|---|---|---|
| 1 | Bypass | New York | Immunology | VALID |
| 2 | Beta-Repair | Los Angeles | Pulmonology | VALID |
| 3 | Gamma-Scan | San Francisco | Cardiology | VALID |
| 4 | Delta-Op | Seattle | Neurology | VALID |
| 5 | Epsilon-Fix | Austin | Orthopedics | VALID |
| 6 | Zeta-TREAT | Atlanta | Oncology | VALID |
| 7 | Eta-PROC | Phoenix | Pediatrics | VALID |
| 8 | Theta-CARE | San Diego | Dermatology | VALID |
| 9 | Iota-SURG | Orlando | Radiology | VALID |
| 10 | Kappa-EXAM | Detroit | Psychiatry | VALID |
| 11 | Lambda-X | Las Vegas | Cardiology | INVALID |
| 12 | Omega-Y | Seattle | Oncology | INVALID |

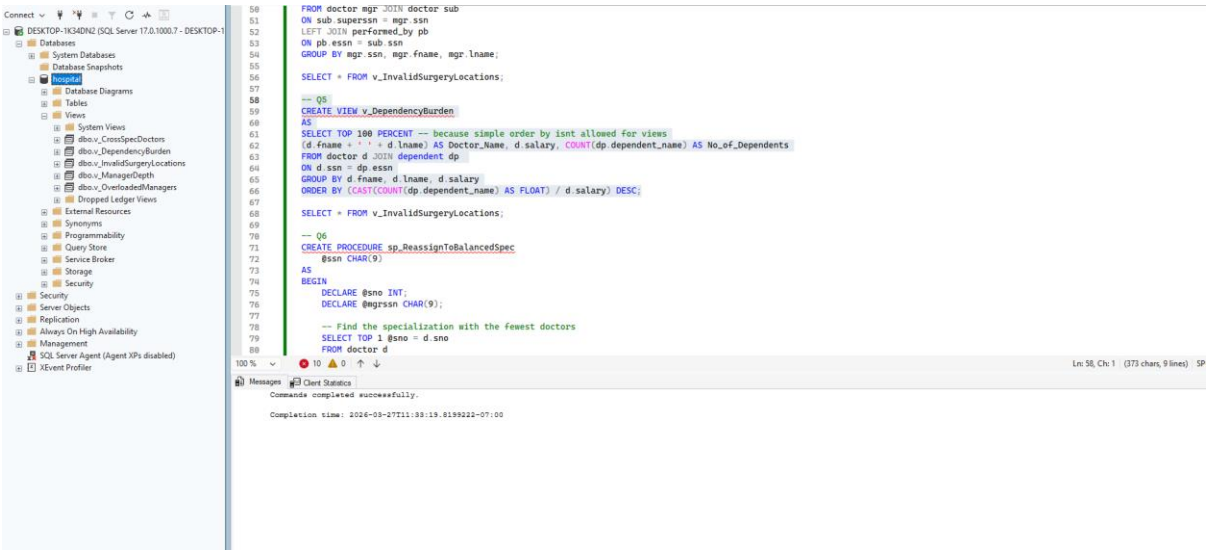
# Q4




| | Surgery_Name | Location | Specialization_Name | Status |
|---|---|---|---|---|
| 1 | Bypass | New York | Immunology | VALID |
| 2 | Beta-Repair | Los Angeles | Pulmonology | VALID |
| 3 | Gamma-Scan | San Francisco | Cardiology | VALID |
| 4 | Delta-Op | Seattle | Neurology | VALID |
| 5 | Epsilon-Fix | Austin | Orthopedics | VALID |
| 6 | Zeta-TREAT | Atlanta | Oncology | VALID |
| 7 | Eta-PROC | Phoenix | Pediatrics | VALID |
| 8 | Theta-CARE | San Diego | Dermatology | VALID |
| 9 | Iota-SURG | Orlando | Radiology | VALID |
| 10 | Kappa-EXAM | Detroit | Psychiatry | VALID |
| 11 | Lambda-X | Las Vegas | Cardiology | INVALID |
| 12 | Omega-Y | Seattle | Oncology | INVALID |

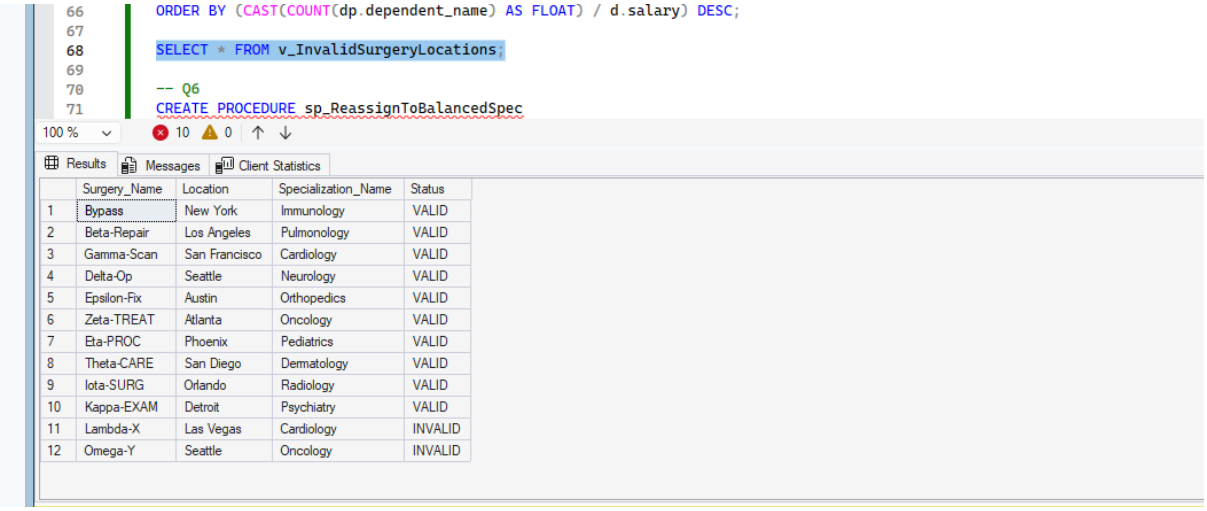
# Q5



```
FROM doctor mgr JOIN doctor sub
ON sub.supersn = mgr.ssn
LEFT JOIN performed_by pb
ON pb.essn = sub.ssn
GROUP BY mgr.ssn, mgr.fname, mgr.lname;

SELECT * FROM v_InvalidSurgeryLocations;

-- Q5
CREATE VIEW v_DependencyBurden
AS
SELECT TOP 100 PERCENT -- because simple order by isnt allowed for views
(d.fname + ' ' + d.lname) AS Doctor_Name, d.salary, COUNT(dp.dependent_name) AS No_of_Dependents
FROM doctor d JOIN dependent dp
ON d.ssn = dp.essn
GROUP BY d.fname, d.lname, d.salary
ORDER BY (CAST(COUNT(dp.dependent_name) AS FLOAT) / d.salary) DESC;

SELECT * FROM v_InvalidSurgeryLocations;

-- Q6
CREATE PROCEDURE sp_ReassignToBalancedSpec
    @ssn CHAR(9)
AS
BEGIN
    DECLARE @sno INT;
    DECLARE @mgrssn CHAR(9);

    -- Find the specialization with the fewest doctors
    SELECT TOP 1 @sno = d.sno
    FROM doctor d
```

Commands completed successfully.

Completion time: 2026-03-27T11:33:19.8199222-07:00



```
ORDER BY (CAST(COUNT(dp.dependent_name) AS FLOAT) / d.salary) DESC;

SELECT * FROM v_InvalidSurgeryLocations;

-- Q6
CREATE PROCEDURE sp_ReassignToBalancedSpec
```

100 % 10 0

Results | Messages | Client Statistics

| | Surgery_Name | Location | Specialization_Name | Status |
|---|---|---|---|---|
| 1 | Bypass | New York | Immunology | VALID |
| 2 | Beta-Repair | Los Angeles | Pulmonology | VALID |
| 3 | Gamma-Scan | San Francisco | Cardiology | VALID |
| 4 | Delta-Op | Seattle | Neurology | VALID |
| 5 | Epsilon-Fix | Austin | Orthopedics | VALID |
| 6 | Zeta-TREAT | Atlanta | Oncology | VALID |
| 7 | Eta-PROC | Phoenix | Pediatrics | VALID |
| 8 | Theta-CARE | San Diego | Dermatology | VALID |
| 9 | Iota-SURG | Orlando | Radiology | VALID |
| 10 | Kappa-EXAM | Detroit | Psychiatry | VALID |
| 11 | Lambda-X | Las Vegas | Cardiology | INVALID |
| 12 | Omega-Y | Seattle | Oncology | INVALID |

# Q6

Solution.sql...ad Raza (90))

```sql
SELECT * FROM v_InvalidSurgeryLocations;

-- Q6
CREATE PROCEDURE sp_ReassignToBalancedSpec
    @ssn CHAR(9)
AS
BEGIN
    DECLARE @sno INT;
    DECLARE @mgrssn CHAR(9);

    -- Find the specialization with the fewest doctors
    SELECT TOP 1 @sno = d.sno
    FROM doctor d
    GROUP BY d.sno
    ORDER BY COUNT(DISTINCT d.ssn) ASC;

    -- Get the manager of that specific specialization
    SELECT @mgrssn = mgrssn
    FROM specialization
    WHERE snumber = @sno;

    -- Update the doctor's record
    UPDATE doctor
    SET sno = @sno,
        superssn = @mgrssn
    WHERE ssn = @ssn;
END;

EXEC sp_ReassignToBalancedSpec @ssn = '123456789';

-- Q7
```

100 % | 10 | 0 | Ln: 70, Ch: 1 | (576 chars, 25 lines) | SPC | CRLF | U

Messages | Client Statistics

```
Commands completed successfully.

Completion time: 2026-03-27T11:33:57.4461158-07:00
```

```sql
SELECT * FROM doctor WHERE ssn = 123456789;
EXEC sp_ReassignToBalancedSpec @ssn = 123456789;
SELECT * FROM doctor WHERE ssn = 123456789;

-- Q7
-- I have learnt to use Cursor which acts like an iterator
```

100 % | 10 | 0 | Ln: 96, Ch: 1 | (140 chars, 3 line

Results | Messages | Client Statistics

| | fname | minit | lname | ssn | bdate | address | sex | salary | superssn | sno |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | John | D | Smith | 123456789 | 1980-01-01 | 123 Main St | M | 50000.00 | NULL | 1 |

| | fname | minit | lname | ssn | bdate | address | sex | salary | superssn | sno |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | John | D | Smith | 123456789 | 1980-01-01 | 123 Main St | M | 50000.00 | 123456789 | 1 |

# Q7

Solution.sql...ad Raza (90))

```sql
SELECT * FROM doctor WHERE ssn = 123456789;

-- Q7
-- I have learnt to use Cursor which acts like an iterator
-- to fetch each record one by one to perform operations
-- I fully understand this code and its working
CREATE PROCEDURE sp_RedistributeSurgeryHours
AS
BEGIN
    -- Variables to hold data for the "overloaded" doctor we are currently processing
    DECLARE @essn CHAR(9);
    DECLARE @sno INT;
    DECLARE @hours DECIMAL(4,1);
    DECLARE @reduced_time DECIMAL(4,1);
    DECLARE @other_doctors_count INT;

    -- A Cursor is like a 'foreach' loop for SQL rows
    DECLARE surgery_cursor CURSOR FOR
        SELECT essn, sno, hours
        FROM performed_by
        WHERE hours > 30;

    OPEN surgery_cursor;
    FETCH NEXT FROM surgery_cursor INTO @essn, @sno, @hours;

    WHILE @@FETCH_STATUS = 0
    BEGIN
        -- Calculate the 20% "burden" to move
        SET @reduced_time = @hours * 0.20;

        -- Count how many OTHER doctors are on this specific surgery
        SELECT @other_doctors_count = COUNT(*)
        FROM performed_by
        WHERE sno = @sno AND essn != @essn;

        -- Only redistribute if there is actually someone else to give it to
        IF @other_doctors_count > 0
        BEGIN
            -- Reduce the original doctor's hours
            UPDATE performed_by
            SET hours = hours - @reduced_time
            WHERE essn = @essn AND sno = @sno;

            -- Distribute the time equally to the others
            UPDATE performed_by
            SET hours = hours + (@reduced_time / @other_doctors_count)
            WHERE sno = @sno AND essn != @essn;
        END

        FETCH NEXT FROM surgery_cursor INTO @essn, @sno, @hours;
    END;

    CLOSE surgery_cursor;
    DEALLOCATE surgery_cursor;
END;

EXEC sp_RedistributeSurgeryHours;
```

75 % 10 0 Ln: 152, Ch: 5 (1770 chars, 53 lines) SPC CRL

Messages | Client Statistics

```
Commands completed successfully.

Completion time: 2026-03-27T11:26:40.2838898-07:00
```

75 % No issues found Ln: 4, Ch: 1 TABS MIXED UTF-8

Query executed successfully. DESKTOP-1K34DN2 (17.0 RTM) DESKTOP-1K34DN2\Asjad ... hospital 00:00:00 Row: 1, Col: 1

```sql
    DEALLOCATE surgery_cursor;
END;

SELECT * FROM performed_by;
EXEC sp_RedistributeSurgeryHours;
SELECT * FROM performed_by;
```

100 % 10 0 Ln: 154

Results | Messages | Client Statistics

| | essn | sno | hours |
|---|---|---|---|
| 1 | 123456789 | 1 | 20.5 |
| 2 | 123456789 | 2 | 10.0 |
| 3 | 159753486 | 6 | 15.0 |
| 4 | 321654987 | 5 | 40.0 |
| 5 | 456123789 | 3 | 25.0 |
| 6 | 654852963 | 10 | 33.0 |
| 7 | 753951486 | 8 | 18.0 |
| 8 | 789456123 | 4 | 35.0 |
| 9 | 852741963 | 9 | 28.5 |
| 10 | 951357486 | 6 | 10.0 |
| 11 | 951357486 | 7 | 22.0 |
| 12 | 987654321 | 2 | 30.0 |
| 13 | 987654321 | 4 | 15.5 |

| | essn | sno | hours |
|---|---|---|---|
| 1 | 123456789 | 1 | 20.5 |
| 2 | 123456789 | 2 | 10.0 |
| 3 | 159753486 | 6 | 15.0 |
| 4 | 321654987 | 5 | 40.0 |
| 5 | 456123789 | 3 | 25.0 |
| 6 | 654852963 | 10 | 33.0 |
| 7 | 753951486 | 8 | 18.0 |
| 8 | 789456123 | 4 | 28.0 |
| 9 | 852741963 | 9 | 28.5 |
| 10 | 951357486 | 6 | 10.0 |
| 11 | 951357486 | 7 | 22.0 |
| 12 | 987654321 | 2 | 30.0 |
| 13 | 987654321 | 4 | 22.5 |

# Q8

```sql
SELECT * FROM performed_by;
EXEC sp_RedistributeSurgeryHours;
SELECT * FROM performed_by;

-- Q8
CREATE PROCEDURE sp_FixSupervisorCycles
AS
BEGIN
    DECLARE @ssn CHAR(9);

    -- Select doctors who are their own direct supervisor
    DECLARE ssn_cursor CURSOR FOR
        SELECT ssn
        FROM doctor
        WHERE ssn = superssn;

    OPEN ssn_cursor;

    FETCH NEXT FROM ssn_cursor INTO @ssn;

    WHILE @@FETCH_STATUS = 0
    BEGIN
        UPDATE doctor
        SET superssn = NULL
        WHERE ssn = @ssn;

        FETCH NEXT FROM ssn_cursor INTO @ssn;
    END;

    CLOSE ssn_cursor;
    DEALLOCATE ssn_cursor;
END;

EXEC sp_FixSupervisorCycles
```

Messages | Client Statistics

```
Commands completed successfully.

Completion time: 2026-03-27T11:38:45.7004427-07:00
```

```sql
    DEALLOCATE ssn_cursor;
END;

SELECT * FROM doctor;
EXEC sp_FixSupervisorCycles
SELECT * FROM doctor;

-- Q9
CREATE PROCEDURE sp_ReplaceManager
    @snumber INT,
    @new_mgrssn CHAR(9)
AS
```

Results | Messages | Client Statistics

| | fname | minit | lname | ssn | bdate | address | sex | salary | superssn | sno |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Tom | K | Holland | 112233445 | 1993-11-11 | 789 Cedar St | M | 57000.00 | 456123789 | 2 |
| 2 | John | D | Smith | 123456789 | 1980-01-01 | 123 Main St | M | 50000.00 | 123456789 | 1 |
| 3 | Sara | F | Wilson | 159753486 | 1992-06-06 | 987 Spruce St | F | 47000.00 | 456123789 | 6 |
| 4 | Lucy | L | Miller | 223344556 | 1996-12-12 | 999 Birch St | F | 60000.00 | 987654321 | 5 |
| 5 | Chris | E | Davis | 321654987 | 1988-05-05 | 654 Maple St | M | 65000.00 | 987654321 | 5 |
| 6 | Emily | B | Jones | 456123789 | 1985-03-03 | 789 Oak St | F | 55000.00 | 123456789 | 3 |
| 7 | Olivia | J | Moore | 654852963 | 1994-10-10 | 579 Chestnut St | F | 62000.00 | 753951486 | 10 |
| 8 | Sop... | H | Ander... | 753951486 | 1991-08-08 | 864 Cedar St | F | 48000.00 | 951357486 | 8 |
| 9 | Mic... | C | Brown | 789456123 | 1975-04-04 | 321 Pine St | M | 75000.00 | 987654321 | 4 |
| 10 | Ethan | I | Thom... | 852741963 | 1995-09-09 | 246 Walnut St | M | 56000.00 | 753951486 | 9 |
| 11 | David | G | Taylor | 951357486 | 1983-07-07 | 135 Birch St | M | 52000.00 | 753951486 | 7 |
| 12 | Jane | A | Doe | 987654321 | 1990-02-02 | 456 Elm St | F | 60000.00 | 123456789 | 2 |

| | fname | minit | lname | ssn | bdate | address | sex | salary | superssn | sno |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Tom | K | Holland | 112233445 | 1993-11-11 | 789 Cedar St | M | 57000.00 | 456123789 | 2 |
| 2 | John | D | Smith | 123456789 | 1980-01-01 | 123 Main St | M | 50000.00 | NULL | 1 |
| 3 | Sara | F | Wilson | 159753486 | 1992-06-06 | 987 Spruc... | F | 47000.00 | 456123789 | 6 |
| 4 | Lucy | L | Miller | 223344556 | 1996-12-12 | 999 Birch St | F | 60000.00 | 987654321 | 5 |
| 5 | Chris | E | Davis | 321654987 | 1988-05-05 | 654 Maple St | M | 65000.00 | 987654321 | 5 |
| 6 | Emily | B | Jones | 456123789 | 1985-03-03 | 789 Oak St | F | 55000.00 | 123456789 | 3 |
| 7 | Olivia | J | Moore | 654852963 | 1994-10-10 | 579 Chestn... | F | 62000.00 | 753951486 | 10 |
| 8 | Sop... | H | Ander... | 753951486 | 1991-08-08 | 864 Cedar St | F | 48000.00 | 951357486 | 8 |
| 9 | Mic... | C | Brown | 789456123 | 1975-04-04 | 321 Pine St | M | 75000.00 | 987654321 | 4 |
| 10 | Ethan | I | Thom... | 852741963 | 1995-09-09 | 246 Walnu... | M | 56000.00 | 753951486 | 9 |
| 11 | David | G | Taylor | 951357486 | 1983-07-07 | 135 Birch St | M | 52000.00 | 753951486 | 7 |
| 12 | Jane | A | Doe | 987654321 | 1990-02-02 | 456 Elm St | F | 60000.00 | 123456789 | 2 |

Query executed successfully.

# Q9

```sql
SELECT * FROM doctor;
EXEC sp_FixSupervisorCycles
SELECT * FROM doctor;

-- Q9
CREATE PROCEDURE sp_ReplaceManager
    @snumber INT,
    @new_mgrssn CHAR(9)
AS
BEGIN
    DECLARE @old_mgrssn CHAR(9);

    -- Get the current (old) manager SSN for this specialization
    SELECT @old_mgrssn = mgrssn
    FROM specialization
    WHERE snumber = @snumber;

    -- Verify the new manager actually belongs to this specialization
    -- This prevents assigning a Pediatrician to manage Orthopedics, for example.
    IF EXISTS (SELECT 1 FROM doctor WHERE ssn = @new_mgrssn AND sno = @snumber)
    BEGIN
        -- Update the Specialization table
        UPDATE specialization
        SET mgrssn = @new_mgrssn,
            mgrstartdate = GETDATE()
        WHERE snumber = @snumber;

        -- Update the Doctor table (Subordinates)
        -- Update superssn only for doctors in this specific specialization
        -- who used to report to the old manager.
        UPDATE doctor
        SET superssn = @new_mgrssn
        WHERE sno = @snumber
          AND superssn = @old_mgrssn;
    END
END;
```

```
Commands completed successfully.

Completion time: 2026-03-27T11:39:52.3745768-07:00
```

Query executed successfully.

```sql
END;

SELECT * FROM specialization;
EXEC sp_ReplaceManager @snumber = 2, @new_mgrssn = 987654321;
SELECT * FROM specialization;
```

| | sname | snumber | mgrssn | mgrstartdate |
|---|---|---|---|---|
| 1 | Immunology | 1 | 123456789 | 2023-01-01 |
| 2 | Pulmonology | 2 | 987654321 | 2023-02-01 |
| 3 | Cardiology | 3 | 456123789 | 2023-03-01 |
| 4 | Neurology | 4 | 789456123 | 2023-04-01 |
| 5 | Orthopedics | 5 | 321654987 | 2023-05-01 |
| 6 | Oncology | 6 | 159753486 | 2023-06-01 |
| 7 | Pediatrics | 7 | 951357486 | 2023-07-01 |
| 8 | Dermatology | 8 | 753951486 | 2023-08-01 |
| 9 | Radiology | 9 | 852741963 | 2023-09-01 |
| 10 | Psychiatry | 10 | 654852963 | 2023-10-01 |
| 11 | Public Rel... | 11 | 951753258 | 2023-11-01 |
| 12 | Strategy | 12 | 123789456 | 2023-12-01 |

| | sname | snumber | mgrssn | mgrstartdate |
|---|---|---|---|---|
| 1 | Immunology | 1 | 123456789 | 2023-01-01 |
| 2 | Pulmonology | 2 | 987654321 | 2026-03-27 |
| 3 | Cardiology | 3 | 456123789 | 2023-03-01 |
| 4 | Neurology | 4 | 789456123 | 2023-04-01 |
| 5 | Orthopedics | 5 | 321654987 | 2023-05-01 |
| 6 | Oncology | 6 | 159753486 | 2023-06-01 |
| 7 | Pediatrics | 7 | 951357486 | 2023-07-01 |
| 8 | Dermatology | 8 | 753951486 | 2023-08-01 |
| 9 | Radiology | 9 | 852741963 | 2023-09-01 |
| 10 | Psychiatry | 10 | 654852963 | 2023-10-01 |
| 11 | Public Rel... | 11 | 951753258 | 2023-11-01 |
| 12 | Strategy | 12 | 123789456 | 2023-12-01 |

Query executed successfully.

# Q10

Solution.sql...ad Raza (90))

```sql
-- Q10
CREATE PROCEDURE sp_DeleteDoctorCascade
    @ssn_to_delete CHAR(9)
AS
BEGIN
    DECLARE @supervisor_ssn CHAR(9);
    DECLARE @specialization_id INT;
    DECLARE @backup_doctor_ssn CHAR(9);

    -- Get the doctor's current supervisor and specialization
    SELECT @supervisor_ssn = superssn, @specialization_id = sno
    FROM doctor
    WHERE ssn = @ssn_to_delete;

    -- Reassign Dependents to the doctor's supervisor
    UPDATE dependent
    SET essn = @supervisor_ssn
    WHERE essn = @ssn_to_delete;

    -- Find a backup doctor for surgeries (someone in the same specialization)
    SELECT TOP 1 @backup_doctor_ssn = ssn
    FROM doctor
    WHERE sno = @specialization_id AND ssn <> @ssn_to_delete;

    IF @backup_doctor_ssn IS NOT NULL
    BEGIN
        UPDATE performed_by
        SET essn = @backup_doctor_ssn, hours = 0 -- reset hours for new doc
        WHERE essn = @ssn_to_delete;
    END

    -- Replace them if they are a Manager of a specialization
    -- I am using the backup doctor found above to take over the department
    IF @backup_doctor_ssn IS NOT NULL
    BEGIN
        UPDATE specialization
        SET mgrssn = @backup_doctor_ssn,
            mgrstartdate = GETDATE()
        WHERE mgrssn = @ssn_to_delete;
    END

    -- 5. Finally delete the doctor now
    DELETE FROM doctor
    WHERE ssn = @ssn_to_delete;

    -- syntax to print a message on screen
    PRINT 'Doctor ' + @ssn_to_delete + ' deleted and responsibilities reassigned.';
END;
```

83 % 10 0

Messages

```
Commands completed successfully.

Completion time: 2026-03-27T11:43:30.3438362-07:00
```

83 % No issues found

Query executed successfully. DESKTOP-1K34DN2 (17.0 RTM) DESKTOP-1K34D

Solution.sql...ad Raza (90))

```sql
    WHERE ssn = @ssn_to_delete;

    -- syntax to print a message on screen
    PRINT 'Doctor ' + @ssn_to_delete + ' deleted and responsibilities reassigned.';
END;

SELECT * FROM doctor;
EXEC sp_DeleteDoctorCascade @ssn_to_delete = 112233445;
SELECT * FROM doctor;
```

110 % 10 0

Results | Messages

| | fname | minit | lname | ssn | bdate | address | sex | salary | superssn | sno |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Tom | K | Holland | 112233445 | 1993-11-11 | 789 Cedar St | M | 57000.00 | 456123789 | 2 |
| 2 | John | D | Smith | 123456789 | 1980-01-01 | 123 Main St | M | 50000.00 | NULL | 1 |
| 3 | Sara | F | Wilson | 159753486 | 1992-06-06 | 987 Spruce St | F | 47000.00 | 456123789 | 6 |
| 4 | Lucy | L | Miller | 223344556 | 1996-12-12 | 999 Birch St | F | 60000.00 | 987654321 | 5 |
| 5 | Chris | E | Davis | 321654987 | 1988-05-05 | 654 Maple St | M | 65000.00 | 987654321 | 5 |
| 6 | Emily | B | Jones | 456123789 | 1985-03-03 | 789 Oak St | F | 55000.00 | 123456789 | 3 |
| 7 | Olivia | J | Moore | 654852963 | 1994-10-10 | 579 Chestnut St | F | 62000.00 | 753951486 | 10 |
| 8 | Sophia | H | Anderson | 753951486 | 1991-08-08 | 864 Cedar St | F | 48000.00 | 951357486 | 8 |
| 9 | Michael | C | Brown | 789456123 | 1975-04-04 | 321 Pine St | M | 75000.00 | 987654321 | 4 |
| 10 | Ethan | I | Thomas | 852741963 | 1995-09-09 | 246 Walnut St | M | 56000.00 | 753951486 | 9 |
| 11 | David | G | Taylor | 951357486 | 1983-07-07 | 135 Birch St | M | 52000.00 | 753951486 | 7 |
| 12 | Jane | A | Doe | 987654321 | 1990-02-02 | 456 Elm St | F | 60000.00 | 123456789 | 2 |

| | fname | minit | lname | ssn | bdate | address | sex | salary | superssn | sno |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | John | D | Smith | 123456789 | 1980-01-01 | 123 Main St | M | 50000.00 | NULL | 1 |
| 2 | Sara | F | Wilson | 159753486 | 1992-06-06 | 987 Spruce St | F | 47000.00 | 456123789 | 6 |
| 3 | Lucy | L | Miller | 223344556 | 1996-12-12 | 999 Birch St | F | 60000.00 | 987654321 | 5 |
| 4 | Chris | E | Davis | 321654987 | 1988-05-05 | 654 Maple St | M | 65000.00 | 987654321 | 5 |
| 5 | Emily | B | Jones | 456123789 | 1985-03-03 | 789 Oak St | F | 55000.00 | 123456789 | 3 |
| 6 | Olivia | J | Moore | 654852963 | 1994-10-10 | 579 Chestnut St | F | 62000.00 | 753951486 | 10 |
| 7 | Sop... | H | And... | 753951486 | 1991-08-08 | 864 Cedar St | F | 48000.00 | 951357486 | 8 |
| 8 | Mic... | C | Brown | 789456123 | 1975-04-04 | 321 Pine St | M | 75000.00 | 987654321 | 4 |
| 9 | Ethan | I | Tho... | 852741963 | 1995-09-09 | 246 Walnut St | M | 56000.00 | 753951486 | 9 |
| 10 | David | G | Taylor | 951357486 | 1983-07-07 | 135 Birch St | M | 52000.00 | 753951486 | 7 |
| 11 | Jane | A | Doe | 987654321 | 1990-02-02 | 456 Elm St | F | 60000.00 | 123456789 | 2 |

Query executed successfully. DESKTOP-1K34DN2 (17.0 RTM) DESKTOP-1K34D